

Skills Summary

Three Kinds of Evidence About a Limit

Use this reference while completing your worksheet or when studying.

The limit is where the outputs are heading, not where they land. $\lim_{x \rightarrow c} f(x) = L$ says the outputs get and stay arbitrarily close to L as x closes in on c from both sides. The value $f(c)$ is a separate question.

Two-sided test: the limit exists exactly when $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$.

Continuity at c : $f(c)$ is defined, the limit exists, and the two are equal. Fail the third and you have a *removable* discontinuity (a hole); fail the second and you have a *jump*.

1. Symbolic: factor, cancel, then substitute

Substituting first is always the right opening move. If it returns a number, that is the limit. If it returns $\frac{0}{0}$, that is not a verdict — it says numerator and denominator share the factor $x - c$. Cancel it and substitute again. Useful factorings: $x^2 - a^2 = (x - a)(x + a)$ and $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$. For a compound fraction, combine the inner fractions first.

Example: Find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

Step 1. Substituting gives $\frac{0}{0}$, so the factor $x - 4$ must be sitting in both parts — factor rather than conclude anything.

Step 2. $x^2 - 16 = (x - 4)(x + 4)$, so for $x \neq 4$ the expression is just $x + 4$, and substituting gives $4 + 4$.

The limit is **8**.

Try it: Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

101 :pæð

2. Numerical: close in from both sides

Pick inputs on *both* sides of c and step them closer — for $c = 1$, use 0.9, 0.99 and 1.1, 1.01. If the two columns settle on the same number, that number is your estimate. A table can only suggest a limit, so always follow it with algebra. If the two sides settle on *different* numbers, the two-sided limit does not exist, and the table has told you something the algebra would have to work for.

Example: Estimate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$ numerically.

Step 1. The expression is undefined at 1, so no single substitution says anything — approach it instead.

Step 2. At $x = 0.99$ the value is **1.99**, and at $x = 1.01$ it is **2.01**.

The two sides squeeze the same number, so the estimate is **2** — and cancelling $x - 1$ confirms it exactly.

Try it: Build a short table on each side of $x = 2$ for $\frac{x^2 - 4}{x - 2}$ and estimate the limit.

check: 4

3. Graphical: read each side separately

Trace the curve in from the left, then from the right, and read the *height* each side is heading for. Open circle: that point is excluded but the height still counts for the limit. Filled dot: that is $f(c)$, and it counts for continuity only. When a graph is given by branch formulas, confirm each side by substituting into the branch that side lives on.

Example: $h(x) = 2x - 1$ for $x < 3$ and $h(x) = x + 3$ for $x > 3$. Find $\lim_{x \rightarrow 3^-} h(x)$.

Step 1. A one-sided limit only ever uses the branch on that side, so the right-hand rule is irrelevant here.

Step 2. To the left of 3 the rule is $2x - 1$, which is continuous, so substitute $x = 3$ into it: $6 - 1$.

The left-hand limit is **5**.

Try it: For the same h , find $\lim_{x \rightarrow 3^+} h(x)$. Do the two one-sided limits agree?

check: 9

(!) Watch out: a filled dot is not the limit — at a removable discontinuity the curve heads for the hole, not for the stray point above it. And $\frac{0}{0}$ is never an answer: it is the signal to factor, rationalize, or combine fractions and try again.